

Examiner reports

Q1.

The trapezium rule was again generally well understood with very little evidence of candidates mixing up 'ordinates' and 'strips'. Fewer candidates than normal failed to give their final answer to the required number of significant figures although there was some evidence of premature approximation/wrong rounding in a minority of solutions.

Part (b), as expected, caused more problems. Generally weaker and average candidates did not understand what was required and many better candidates failed to use brackets

sensibly and gave the expression for $f(x)$ as ' $\sqrt{2x^3 + 1}$ ', for which they were awarded 1 mark.

Q2.

In part (a), many candidates gave the correct value for where the curve met the y -axis, but a significant minority drew no part of the graph in the second quadrant, their graphs stopping at the point (0, 1). Many candidates scored at least half marks for their descriptions of the geometrical transformations in part (b). The most common wrong answer in part (b)(i) was "a stretch of scale factor 2 parallel to the y -axis", and in part (b)(ii) the direction of the translation vector caused the most problems.

Part (c)(i) was poorly answered, with many candidates failing to give sufficient justification for the printed answer. A very common error was to write 9^x as $3y$ and yet still reach the correct factorised quadratic equation.

Although part (c)(ii) was answered better, the majority of candidates failed to gain all the four marks. The most common wrong approach was to take logarithms of each term in the given equation. Those candidates applying this incorrect method had clearly not seen the implication of the "Hence". Those candidates who had seen the link and started with $3^x = 1$ and $3^x = 2$ generally scored all four marks.

Q3.

This question was not very well answered, with few candidates gaining all 4 marks. The most common problem was an inability to give the appropriate scale factor when stretching in the x -axis.

Q4.

This question was not answered very well in general. The fractional values of p and q caused problems.

In part (a), although many found the value of p correctly, far fewer obtained the correct value of q . Once more arithmetic errors occurred, particularly when calculating $4 - \frac{9}{4}$.

Very few candidates understood what was required for part (b). Many hedged their bets and wrote their answer in coordinate form, (1.5, 1.75), whilst many others thought the minimum value was 1.5.

In part (c), many earned a mark for the word “translation”, provided it was not combined with another transformation. However, the wrong vector was very common; few recognised the link with part (a) and used 3 and 4 instead of their values of p and q .

Q5.

Candidates generally applied the trapezium rule correctly and gave the final answer to the given degree of accuracy. Many candidates failed to gain full marks in part (b)(ii) as their diagrams did not show four trapeziums. There were also a significant number of candidates who either failed to draw any diagram or who drew the sloping sides of their trapeziums under the curve. Most candidates realised that the geometrical transformation was a stretch, but the wrong answer ‘stretch parallel to the y -axis, scale factor 3’ appeared almost as frequently as the correct answer. A high proportion of candidates showed that they were able to use logarithms to solve the equation $6^{3x} = 84$ but some only gave their answer to two decimal places. The final part of the question was poorly answered, although a minority of candidates did score some credit for partially correct answers, for example, $f(x) = 6^{x+1} - 2$.

Q6.

Part (a) was usually B1B0 with the shape being correct but the coordinates of the intercepts on the axes were often incorrect.

In part (b)(i) many fully correct answers were seen and most candidates earned partial credit. Mistakes usually occurred where candidates started with the translation and gave the combined vector. Where candidates started with the stretch they were far more successful.

Part (b)(ii) was very well answered with many candidates earning full marks.

Q7.

Most candidates were able to give a correct value in part (a) but a significant minority did not give any second value, despite the hint in the question: “answers”. A common wrong answer was “4.407”. In part (b)(i), some candidates mixed up the coordinates and in part (b)(ii) “ $(\pi + a)$ ” was the usual wrong answer. The description of the required transformation in part (c) was answered better this year.

Although some excellent solutions were seen to part (d), these were very much the exception. The vast majority of candidates could not deal correctly with the $2x$ beyond

finding the first solution, $x = \frac{2\pi}{5}$.

Q8.

This question was well answered by the majority of candidates. Common incorrect answers involved the use of 3 rather than $1/3$ or naming the y -axis rather than the x -axis when referring to the stretch. The answer of “squash – scale factor 3 – (in the x -direction)” did occur on more occasions than had been expected. The word “transformation” was often used instead of “translation” and the vector for the translation was often described in words rather than stated as a vector.

Q9.

Parts (a) and (b) were usually well done, but in part (b), although many completely correct answers were seen, there were also many diagrams where part (a) had been translated 4 units along the x or y axis and W-shape diagrams were also popular. Dealing with the modulus in part (c) proved to be challenging, although there were many fully correct

answers. Candidates appeared to be more successful with finding $x = \frac{4}{3}$ than $x = -4$. A

surprising number of candidates followed $3x = 4$ by $x = \frac{3}{4}$.

Those candidates who were successful in part (c) were usually successful in part (d) but correct values with incorrect inequality signs was a common error.

Q10.

Part (a)(i) The completion of the square was done successfully by most candidates, although occasionally $+44$ was seen instead of -6 for q .

Part (a)(ii) Most candidates were able to write down the correct minimum point, although some wrote $(5, -6)$ as the vertex. A few chose to use differentiation but often made arithmetic slips in finding the coordinates of the stationary point.

Part (a)(iii) Although there were correct answers for the equation, the term “line of symmetry” was not well understood by many; typical wrong answers were $y = -6$, the y -axis and even $y = -x^2 + 10x + 19$ or other quadratic curves.

Part (a)(iv) The more able candidates earned full marks here. The term **translation** was required but generally the wrong word was used or it was accompanied by another transformation such as a stretch. The most common (but incorrect) vector stated was

$$\begin{bmatrix} 10 \\ 19 \end{bmatrix}$$

Part (b) There were a number of complete correct solutions here. The errors that did occur usually stemmed from sign slips in rearranging the equations. Some candidates found the x -coordinates and made no attempt at the y -coordinates. A few candidates wrote down the coordinates of at least one point without any working.

Q11.

Candidates usually have difficulty with trigonometry and this session was no exception. Sketching of the graph of $y = \tan x$ was generally not done well and perhaps illustrated the over-reliance on calculators for graphical work. Surprisingly the first branch, between $x = 0^\circ$ and $x = 90^\circ$, was missing in a significant number of sketches. In part (b) candidates were expected to be able to just write down the two values by first noting by inspection that $x = 61$ was a solution. Instead many gave the incorrect solutions $\tan 61$ and $\tan 241$.

Candidates' solutions to show that $\tan \theta = -1$ in part (c)(i) were often unconvincing. The examiners were generally looking for a division by $\cos \theta$ before or after a rearrangement.

Part (c)(ii) starts with the word ‘Hence’ so those candidates who only gave the answers 155° and 335° gained no credit. Those candidates who spotted the link and indicated ‘ $\tan(x - 20^\circ) = -1$ ’ generally went on to gain full marks, although some others subtracted (instead of added) 20 from their answers to part (c)(i). A common error was to start by writing ‘ $\sin(x - 20) + \cos(x - 20) = \sin x - 20 \sin + \cos x - \cos 20$ ’ from which candidates never recovered even after a page of working.

Many candidates gained at least 1 mark for their description of the geometrical transformation in part (d). The final part of the question was answered well by the above average candidates but some left their answer as $f(4x)$ which was not awarded the mark. A substantial number of candidates gave the answer as $4f(x)$.

Q12.

Most candidates were able to gain some marks in part (a). In (i) the common wrong answer was a stretch in the x -direction with scale factor $1/2$. In (ii) most gave a correct reflection and in (iii) many gave the correct vector or its equivalent but still some candidates are not using the correct word ‘translation’. As previously stated, ‘tr’ was not accepted as a replacement for ‘translation’.

Part (b) was not answered well. The usual common error, illustrated by ‘ $\theta - 30 = 44.4$, $\theta = 74.4$ or $\theta = 180 - 74.4$ ’ was seen too frequently. It was not uncommon to see candidates giving answers in all four quadrants. A more surprising error from the better candidates is illustrated by ‘ $\theta - 30 = 44.4$, 135.6 so $\theta = 14.4$, 105.6 ’

Part (c) was answered well but the usual errors $(\cos x + \sin x)^2 = \cos^2 x + \sin^2 x$ and $(\cos x + \sin x)^2 = \cos^2 x + \sin^2 x + \cos x \sin x$ and the ‘trig identity ‘ $\cos x + \sin x = 1$ ’ were all seen.

Q13.

Part (a) Many candidates began by finding the correct values of p and q . A few wrote $(x - 4)^2$ and some added 16 instead of subtracting 16 so $q = 35$ was sometimes seen.

Part (b) Very few chose to consider the expression they had in part (a). Practically all candidates decided to find the discriminant instead but its evaluation was often incorrect. Not everyone quoted the expression for the discriminant, $b^2 - 4ac$, correctly. Some attempted to refer to the fact that the curve was completely above the x -axis but did not, in general, complete their argument.

Part (c) The graphs here were disappointing. Although most drew a quadratic shape, there seemed to be little reference to their part (a) and many just tried to plot a few points. Most were able to state the intercept on the y -axis. However, sometimes the point (0.19) was shown as the minimum point or a straight line intercept. Several curves were drawn only in the first quadrant, regardless of whether the quoted minimum point was (-4.3) or (4.3) .

Part (d) This was not well answered. The term **translation** was required but generally the wrong word was used or it was accompanied by another transformation such as a

stretch. The most common incorrect vector stated was $\begin{bmatrix} 8 \\ -19 \end{bmatrix}$.

Q14.

This question proved to be a good source of marks for many candidates. Most candidates answered both parts of (a) correctly although the usual errors, $3^\circ = 0$ or $3^\circ = 3$ and $3^x = 27 \Rightarrow x = \sqrt[3]{27}$ were seen. The trapezium rule, required in part (b), was generally well understood although some candidates are still mixing up 'ordinates' and 'strips' or failing to use sufficient brackets correctly. Although many candidates scored well in parts (c) there were others in (i) who used the relevant law of logarithms incorrectly or others who failed to indicate that logarithms had been used at all. In (ii) not all candidates realised that a simple substitution of 13 for 3^x in $k = 27 - 3^x$ with evaluation was all that was required. Descriptions of transformations continue to pose a problem for candidates. Many candidates gained partial credit for their sketches but both marks were rarely awarded.

Q15.

This question was answered badly by the majority of candidates. In part (a) the examiners required the word 'Stretch' and either the correct direction or the correct scale factor before awarding either of the two marks. It was not uncommon to see the wrong scale

factor $\frac{1}{2}$ and the wrong direction (parallel to the y -axis). Many weaker candidates who attempted part (b) started by writing the incorrect statement ' $x = \tan^{-1}6$ '. Many better solutions were spoiled by manipulation errors illustrated by

$\frac{1}{2}x = 1.249, 4.3906. \Rightarrow x = 0.6245, 2.1958$. Part (c) was by far the worst answered part on the paper. Only a very few candidates realised that $\cos 9 = 0$ satisfied the given factorised form of the equation. The vast majority multiplied out the brackets and then cancelled the $\cos \theta$ factor. A mark was awarded to a significant number of candidates

for knowledge of the identity ' $\tan \theta = \frac{\sin \theta}{\cos \theta}$ ' in this part of the question.

Q16.

In part (a), the question asked for the coordinates of the marked points on the trigonometrical graph. For each point, examiners expected to be given both coordinates rather than just a non-zero value. Other common errors included answers involving radians and also answers which had the x - and y -coordinates switched. In part (b), candidates were expected to use the word "stretch" and also to give the direction and scale factor of the stretch. As expected, a scale factor of 2 rather than $1/2$ was the usual error.

Relatively few candidates gained all the available marks in part (c). Most obtained two

of the four values, but many missed the value from ' $x = 180 + 180 - \frac{\alpha}{2}$ '. Candidates should also be aware that when asked to give answers to the nearest 0.1° , their earlier working should be done to a greater degree of accuracy. Failure to do this frequently lead to the wrong answer "214.2°".

Q17.

Many candidates incorrectly evaluated $(3^0 + 1)$ as 1 in part (a) although surprisingly they frequently recovered and used the correct answer, 2, in part (b). In general, the trapezium

rule was well understood, although it was relatively common to find an arithmetical slip in evaluating $(3^x + 1)$; the '1' was not added in one of the terms. Candidates should have given the final answer to the degree of accuracy specified in this numerical integration question. Some of the weaker candidates just integrated $(3^x + 1)$ directly as $(3^{x+1} + x)$. The use of a diagram to show that the approximation is an overestimate was well understood, although some used rectangles rather than the required trapeziums and others had the sloping sides of their trapeziums in the wrong positions, that is, below the curve. Part (c) was generally well answered although some weaker candidates tried to take logarithms of each term rather than to rearrange the equation into the form ' $3^x = 4$ ' as a first step. Those candidates who used trial and improvement usually failed to produce an acceptable solution. In part (d), a common wrong expression for $f(x)$ was " $-3^x + 1$ ", although a significant minority did obtain the correct answer " $3^{-x} + 1$ ".

Q18.

The variation between candidates was immense. Some did not appear to have covered this topic in any depth.

Part (a) About two thirds of the candidates found the correct values of p and q but it was alarming to see $(x - 3)^2 - 9 + 16$ become $(x - 3)^2 + 5$ or $(x - 3)^2 + 25$ on so many occasions.

Part (b)(i) Only the confident candidates wrote down $(3, 7)$ for the vertex with no further working. Many used differentiation and found the correct coordinates, but others simply wrote either 3 or 7 as their answer. Quite a few felt that the minimum point was $(-3, 7)$. Common wrong answers from erroneous thinking were $(3, 0)$, $(7, 0)$, $(0, 16)$ or $(7, 16)$.

Part (b)(ii) Almost everyone drew a parabola though a few failed to sketch beyond the minimum point; a few graphs had a maximum point and a few cubic graphs were seen. However the positioning of the quadratic graph was crucial and often bore no connection to their vertex or the correct curve. To gain full marks the parabola had to be correct with the y intercept value of 16 clearly indicated.

Part (b)(iii) Correct curves were generally followed by correct equations for the line of symmetry, but a few gave their answer as $y = 3$. A significant number made no attempt at this part of the question.

Part (c) Full marks here could only be obtained by using the correct term '**translation**'. Using words such as 'shift', 'move' or 'transformation' were not good enough for full marks. The best candidates used a vector, but correct descriptions of how many units the curve had been translated in appropriated directions were of course credited. However, many included a second transformation such as a stretch and this negated the first mark.

Many candidates had incorrect vectors such as $\begin{bmatrix} -3 \\ -7 \end{bmatrix}$ or $\begin{bmatrix} 3 \\ 7 \end{bmatrix}$, or most commonly $\begin{bmatrix} 6 \\ 16 \end{bmatrix}$.

Q19.

This question proved to be a good source of marks for many candidates. In part (a)(i) most candidates quoted the correct value for the power p . Although many candidates produced correct answers for the integration, all the usual errors, as illustrated by

$$\frac{3}{2}x^{1.5}, x^{1.5}, \frac{x^{-0.5}}{-0.5}, \text{ and } \frac{\sqrt{x}^2}{2},$$

were seen. Most candidates stated and used the correct limits in part (a)(iii) but evaluation was sometimes poor. Many of the candidates found a correct equation for the tangent but it was surprising to find a significant minority of these candidates either making no attempt to find the area of triangle AOB or giving the area as a negative value. For full marks in part (c) the examiners were expecting to see the word 'translation'. Some candidates wrote 'tr' but this was not considered to be clear enough to distinguish between 'transformation' and 'translation'. Common wrong answers involved 'stretch' and vectors in the wrong direction. In part (d) there was a significant minority of candidates starting their solution with the incorrect statement $\sqrt{x-1} = \sqrt{x} - \sqrt{1} = \sqrt{x} - 1$ but in general the trapezium rule was well understood and many candidates were able to give the final answer to the correct required degree of accuracy in this numerical integration question.

Q20.

This question discriminated well between candidates. Part (a) showed an alarming lack of the correct use of mathematical language. Candidates had to use the words translation and stretch. Centres must stress this to all their students. There were many correct solutions to part (b) although some candidates missed the final mark by giving their answer as a decimal rather than the exact value that was required. Many candidates scored some marks on the final part but there were few fully correct solutions.



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